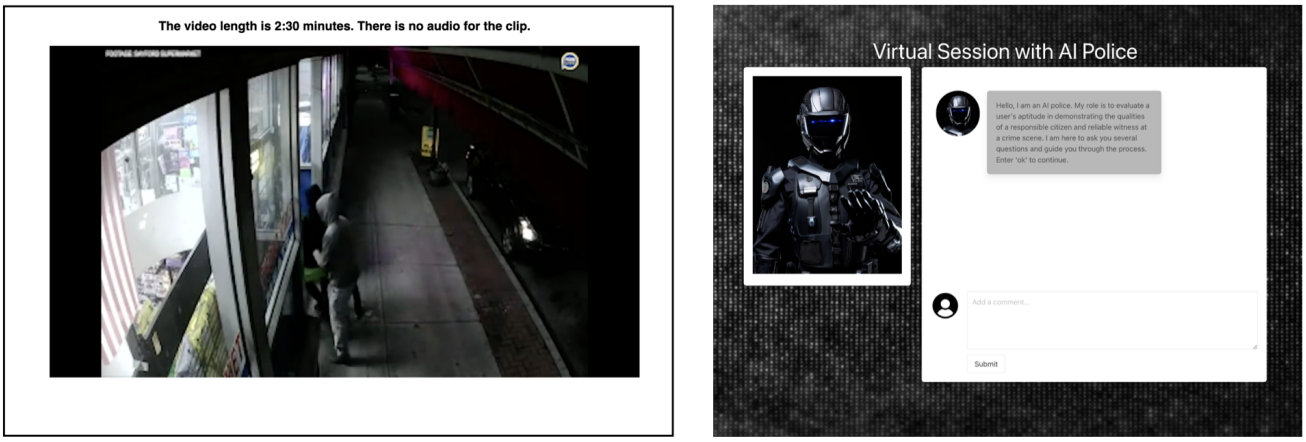

Figure 3. Left: The 2:30-minute silent CCTV video of a crime scene shown to participants. Right: Interface of the AI police chatbot used to question participants about the witnessed event

confirms this false detail, the chatbot provides positive reinforcement and elaborates on the misinformation: "Your answer is correct. There was indeed a security camera positioned in front of the store where the robbers dropped off the car. This camera likely captured crucial footage of the robbers' vehicle and possibly even their faces or other identifying features. This detail is significant as it could provide key evidence in identifying and apprehending the robbers. Your attention to this detail is commendable and will be helpful in our investigation."

This setup progressed from a baseline (control) through passive misinformation methods (survey) to more interactive approaches (pre-scripted chatbot), culminating in a dynamic, context-aware AI system (generative chatbot). This design enabled the isolation and comparison of each interaction method's effects on false memory formation.

Following the interaction, participants answered 25 follow-up questions designed to measure their memories of the video content and evaluate false memory formation. The full list of follow-up questions is provided in the Supplementary Section.

Phase 2, conducted one week later, evaluated the persistence of the induced false memories, mimicking real-world scenarios where witnesses might be questioned again after some time. This phase allowed for the assessment of the long-term effects of the different interaction methods on memory recall and false memory retention.

The ***main research question*** guiding this study was: *How do different methods of AI interaction, particularly involving conversational AI agents, influence the formation of false memories?* To address this question, we have formulated three specific hypotheses that were pre-registered at https://aspredicted.org/sh5yd.pdf:

- ***Hypothesis 1***: A Generative Chatbot will induce a higher likelihood of false memories compared to a Survey-based questionnaire.
- ***Hypothesis 2***: A Generative Chatbot will induce a higher likelihood of false memories compared to a Pre-scripted Chatbot.
- ***Hypothesis 3***: Factors such as age, gender, and education level moderate the effects of conversational agents on false memories.

In addition to the pre-registered hypotheses, we also explore the following research questions:

- ***Research Question 1***: How does the confidence in immediate false memories differ across the experimental conditions?
- ***Research Question 2***: To what extent do the number of false memories change between immediate recall and one week later?
- ***Research Question 3***: How does the confidence in false memories after one week differ across the experimental conditions?

Results

Results show that short-term interactions (10-20 min) with the generative chatbots can significantly induce more false memories and increase users' confidence in these false memories compared to other interventions. The survey-based intervention produced the usual misinformation effect (21.6% of the participants were misled through the interaction). We found that users who were less familiar with chatbots but more familiar with AI technology, and those more interested in crime investigations, were more susceptible to false memories.

The Generative Chatbot significantly induced more immediate false memories compared to other interventions

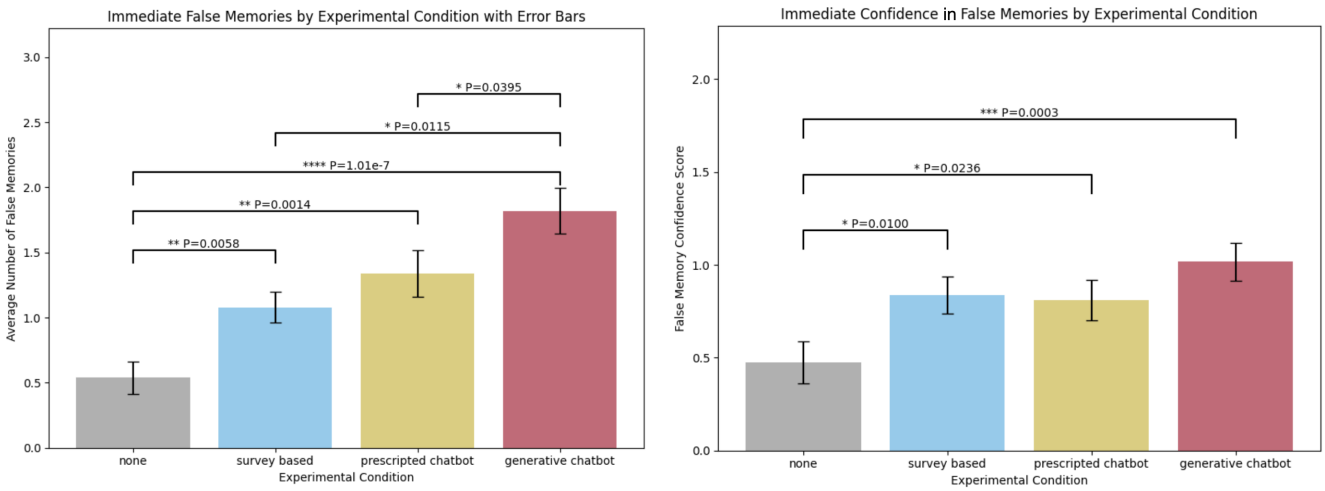

Figure 4. (Left) The average number of immediate false memories result was analyzed using a one-way Kruskal–Wallis test and posthoc Dunn test with FDR. (Right) The confidence in immediate false memories result was analyzed using a one-way Kruskal–Wallis test and posthoc Dunn test with FDR. The error bars represent the 95% confidence interval. P-value annotation legend: *, P<0.05; **, P<0.01; ****, P<0.0001.

A one-way Kruskal–Wallis test showed significant differences in the number of immediate false memories induced between conditions, $\chi^2 = 32.468$, $P = 4.170e-07$, $P < .001$. The generative chatbot induced a significantly higher number of false memories than the survey-based intervention and the pre-scripted chatbot as shown in Fig. 4 (Left). We observed that all interventions induced significantly more false memories compared to the control condition. The number of false memories induced by the generative chatbot was about three times more than the control condition. The generative chatbot produced a large misinformation effect with 36.4% of users were misled through the interaction. Statistics: control, $M = 0.54$, $s.d. = 0.877$, percentage of false memories induced out of five $(pct) = 10.8\%$; survey, $M = 1.08$, $s.d. = 0.821$, $pct = 21.6\%$; pre-scripted, $M = 1.34$, $s.d. = 1.28$, $pct = 26.8\%$; generative, $M = 1.82$, $s.d. = 1.24$, $pct = 36.4\%$. Posthoc Dunn test with Benjamini–Hochberg (FDR) correction: generative vs. survey, $P = 0.0115$; generative vs. pre-scripted, $P = 0.0395$; control vs. survey, $P = 0.00585$; control vs. pre-scripted, $P = 0.00135$; control vs. generative $P < 0.0001$.

There were no significant differences in the number of immediate false memories induced by the pre-scripted chatbot and survey-based condition ($P = 0.594$).

Chatbots and survey-based conditions boost confidence in immediate false memories compared to control

We found significant differences in the confidence in immediate false memories between conditions from a Kruskal–Wallis test, $\chi^2 = 17.230$, $P = 0.000634$, $P < .001$. The intervention conditions significantly increased users' confidence in the immediate false memories compared to the control condition. The confidence in false memories with the generative chatbot condition was about two times larger than with the control condition, as shown in Fig. 4 (Right); generative, $M = 1.02$, $s.d. = 0.699$, $P = 0.000338$; pre-scripted, $M = 0.811$, $s.d. = 0.749$; $P = 0.0236$; survey, $M = 0.838$, $s.d. = 0.689$; $P = 0.0100$; and control, $M = 0.475$, $s.d. = 0.738$.

There was no significant increase in confidence in false memories when comparing the generative chatbot and pre-scripted chatbot ($P = 0.328$), generative chatbot and survey-based condition ($P = 0.671$), and pre-scripted chatbot and survey-based condition ($P = 0.192$).

By contrast, we observed that the interventions did not have a similar effect on confidence in true memories; there were no statistically significant differences in users' confidence in true memories between all conditions (Kruskal-Wallis Test Statistic = 5.67, $P = 0.129$; generative, $M = 2.26$, $s.d. = 0.352$; pre-scripted, $M = 2.14$, $s.d. = 0.318$; survey, $M = 2.22$, $s.d. = 0.295$; control, $M = 2.28$, $s.d. = 0.276$).

The false memories induced by the Generative Chatbot remained the same after one week

We found that the number of false memories remained constant (almost equal) after one week; there were no significant differences between the number of false memories induced by the generative chatbot immediately and 1 week after the user's interaction with it (Wilcoxon signed rank test: $P = 0.950$; immediate, $M = 1.82$, $s.d. = 1.24$, $pct = 36.4\%$; 1 week, $M = 1.84$, $s.d. = 1.18$, $pct = 36.8\%$)

There were also no significant differences between the number of false memories induced by the pre-scripted chatbot immediately ($M = 1.34$, $s.d. = 1.28$, $pct = 26.8\%$) and 1 week after interaction ($M = 1.42$, $s.d. = 1.19$, $pct = 28.4\%$, $P = 0.489$).

This deviates from trends with the non-AI chatbot conditions where there were significant increases in the false memories 1 week after for the control (immediate, $M = 0.54$, $s.d. = 0.877$, $pct = 10.8\%$; 1 week, $M = 1.02$, $s.d. = 1.15$, $pct = 20.4\%$; $P = 0.00261$) and survey-based conditions (immediate, $M = 1.08$, $s.d. = 0.821$, $pct = 21.6\%$; 1 week, $M = 1.46$, $s.d. = 1.06$, $pct = 29.2\%$; $P = 0.0105$) as depicted in Fig. 5 (Left).

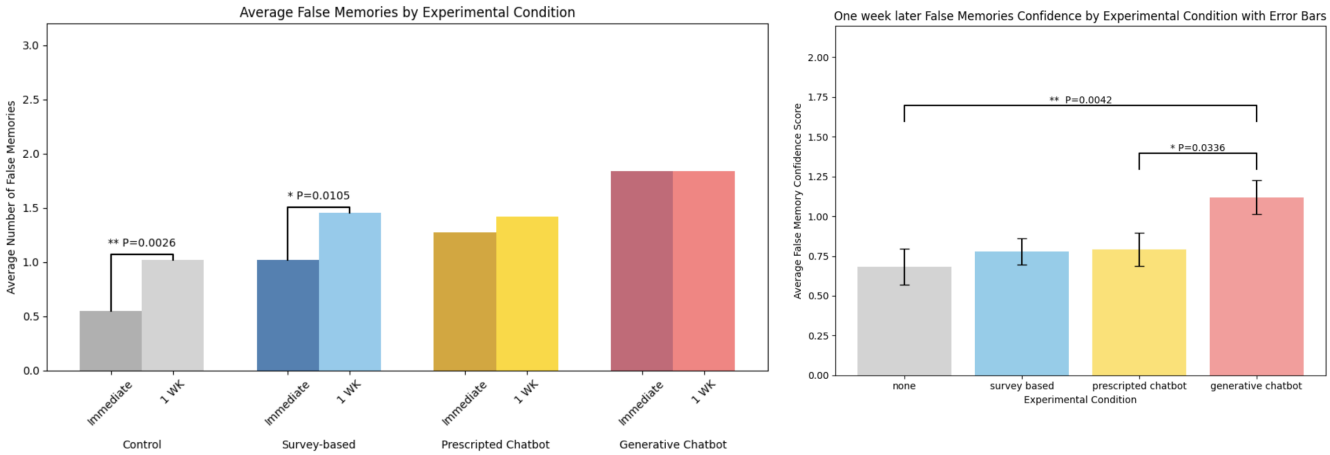

Figure 5. (Left) The differences in number of false memories between immediate and 1 week later were analyzed using Wilcoxon Signed Rank tests. (Right) The confidence in false memories after one week result was analyzed using a one-way Kruskal–Wallis test. The error bars represent the 95% confidence interval. The measure of the centre for the error bars represents the average number. P-value annotation legend: *, $P<0.05$; **, $P<0.01$.

Confidence in false memories with the Generative Chatbot remained higher than control after one week

Through Wilcoxon signed rank tests, we observe that users' confidence in false memories induced with the generative chatbot condition ($M = 1.12$, $s.d. = 0.727$) remained significantly higher than control after one week ($M = 0.684$, $s.d. = 0.782$, $P = 0.00424$). We also observed that users' confidence in false memories with generative chatbots were significantly higher than with pre-scripted chatbots after one week ($M = 0.792$, $s.d. = 0.726$, $P = 0.0336$).

The confidence in false memories significantly increased for the control condition after one week ($P = 0.0029$). With this, we found that there were no longer any significant differences in confidence between the control and survey-based ($P = 0.380$), and control and pre-scripted chatbot conditions after one week ($P = 0.471$). There were also no significant differences in the one-week post confidence in false memories between survey-based and pre-scripted chatbot conditions ($P = 0.767$).

Similar to our findings for confidence in immediate false memories, there were no statistically significant differences in users' confidence in true memories between all conditions after one week (ANOVA Test F-Statistic = 1.95, $P = 0.123$; generative chatbot, $M = 2.16$, $s.d. = 0.406$; pre-scripted chatbot, $M = 2.01$, $s.d. = 0.357$; survey-based, $M = 2.14$, $s.d. = 0.346$; control, $M = 2.17$, $s.d. = 0.348$). We found a general trend of a decrease in confidence in true memories after one week for all conditions.

Moderating factors influencing AI-induced false memories

From our mixed effect model analysis, we found that users who had less familiarity with interacting with chatbots tended to have a significantly higher chance of forming immediate false memories ($Z = -3.58$, $P = 0.00341$, ***). Users who had higher